



LINEAR ALGEBRA

UE19MA251

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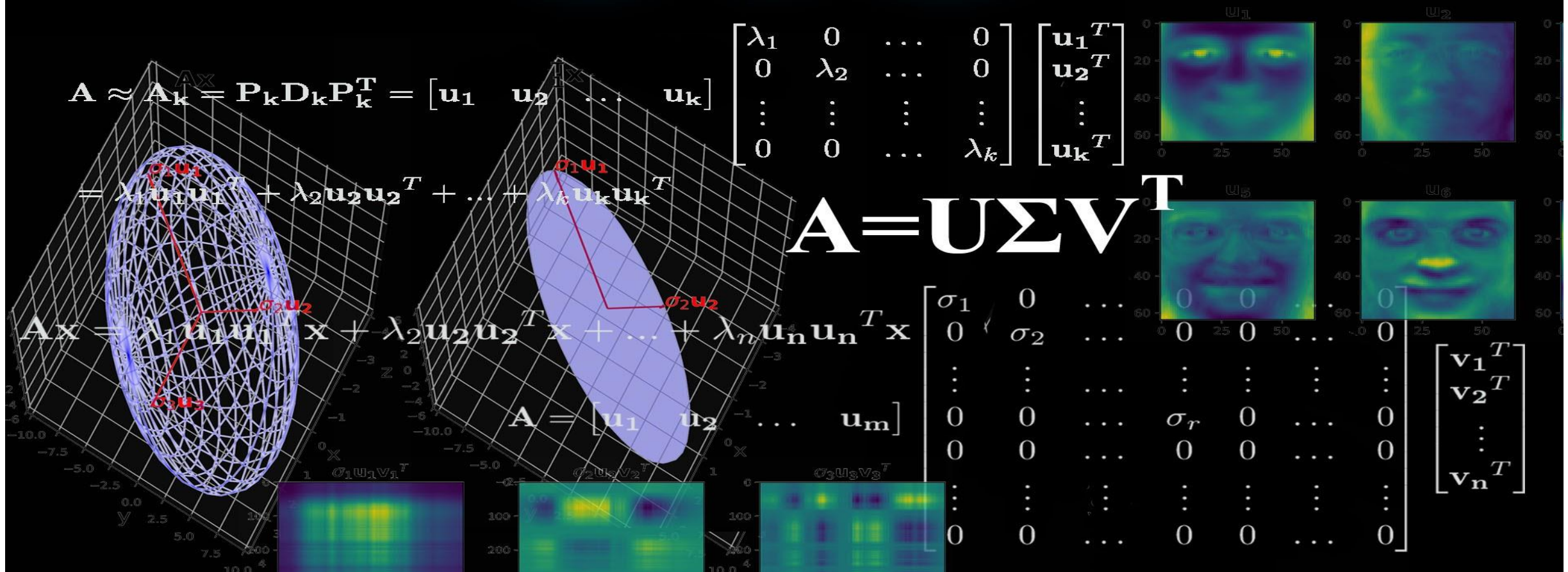


Unit 5 Singular Value Decomposition

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SVD



<https://towardsdatascience.com/understanding-singular-value-decomposition-and-its-application-in-data-science-388a54be95d>

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Importance of the method:



- ☐ Intimately related to the familiar theory of diagonalizing a symmetric matrix.
- ☐ Factorizes a matrix into 3 components.
- ☐ Has interesting algebraic properties.
- ☐ Gives further geometric and theoretical insights.
- ☐ Has many applications to data science.

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Agenda for the chapter



- Tests for positive definiteness
- Positive Definite Matrices and Least Squares
- Semi definite Matrices
- Singular Value Decomposition
- Applications of the SVD.

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Agenda for the class



- Quadratic form
- Examples on quadratic form
- Quadratic form – Going the other way.
- Quadratic forms for a non-symmetric matrix
- Graphs of quadratic forms
- Examples

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Recall



□ If A is a real symmetric matrix of order n by n , then there exists an orthonormal matrix V whose columns are the eigenvectors of A and a diagonal matrix D , having its diagonal entries as the eigenvalues of A , such that : $A = VDV^T$

□ The above process gives the eigenvalue decomposition of the matrix A .

□ The singular value decomposition (SVD) is intimately related to the eigenvalue decomposition.

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Quadratic Form



Any $n \times n$ real symmetric matrix A determines a quadratic form q_A

in n variables by the formula

$$q_A(x_1, \dots, x_n) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j = \mathbf{x}^T A \mathbf{x}.$$

Conversely, given a quadratic form in n variables, its coefficients can be arranged into an $n \times n$ symmetric matrix.

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Quadratic Form

✓ Note : $x^T A x$ is a scalar

$$\therefore x^T A x = (x_1 \ x_2 \ \dots \ x_m) \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1m} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2m} \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mm} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$$

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Quadratic Form

$$\begin{aligned} & x_1(a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m) \\ & + x_2(a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m) \\ & + \vdots \\ & + x_m(a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mm}x_m) \\ & = \sum_{i \leq j}^m a_{ij} x_i x_j \end{aligned}$$

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Quadratic Form – Example 1



Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Compute $\mathbf{x}^T A \mathbf{x}$ for the following matrices.

a. $A = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$

b. $A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$

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Quadratic Form - Example



Solution:

$$\text{a. } \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4x_1 \\ 3x_2 \end{bmatrix} = 4x_1^2 + 3x_2^2.$$

$$\begin{aligned} \text{b. } \mathbf{x}^T \mathbf{A} \mathbf{x} &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3x_1 - 2x_2 \\ -2x_1 + 7x_2 \end{bmatrix} \\ &= x_1(3x_1 - 2x_2) + x_2(-2x_1 + 7x_2) \\ &= 3x_1^2 - 2x_1x_2 - 2x_2x_1 + 7x_2^2 \\ &= 3x_1^2 - 4x_1x_2 + 7x_2^2 \end{aligned}$$

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Quadratic Form – Example 2

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$$

We could rewrite this in the form $Q(x) = 3x_1^2 - 4x_1x_2 + 7x_2^2$.

$$= x^T Ax,$$

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Quadratic Form – Going the other way



Question: Is any function of the form $Q(x) = ax_1^2 + bx_1x_2 + cx_2^2$ a quadratic form?

Answer: Yes. Set $A = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}$.

Question: What about $Q(x) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$?

Answer: Set $A = \begin{bmatrix} 5 & -1/2 & 0 \\ -1/2 & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix}$.

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Quadratic Form – What if A isn't symmetric?



If A isn't symmetric, the function $Q(x) = x^T A x$ is still a quadratic form:

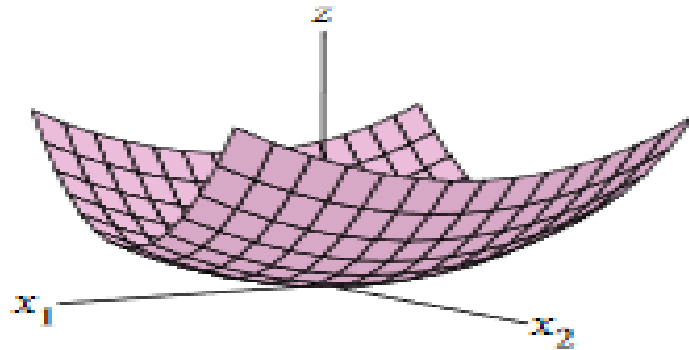
Define $\hat{A} = \frac{A + A^T}{2}$ then

$$\begin{aligned} x^T \hat{A} x &= x^T \left(\frac{A + A^T}{2} \right) x \\ &= \frac{1}{2} \left(x^T A x + x^T A^T x \right) \\ &= \frac{1}{2} \left(x^T A x + x^T A x \right) = x^T A x. \end{aligned}$$

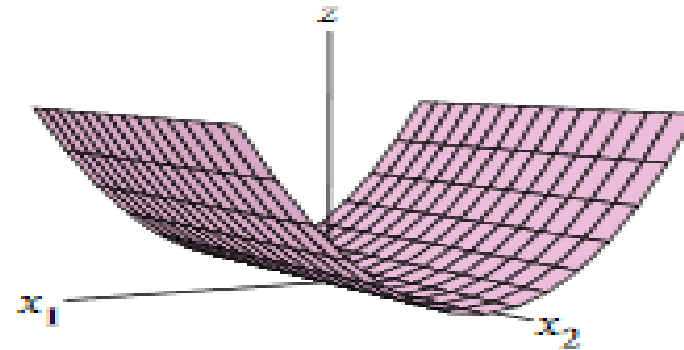
Because of this, it is safe to assume that A is symmetric when we examine quadratic forms.

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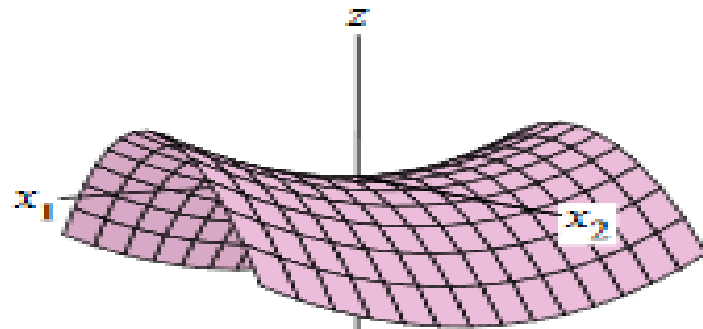
Graphs of Quadratic forms



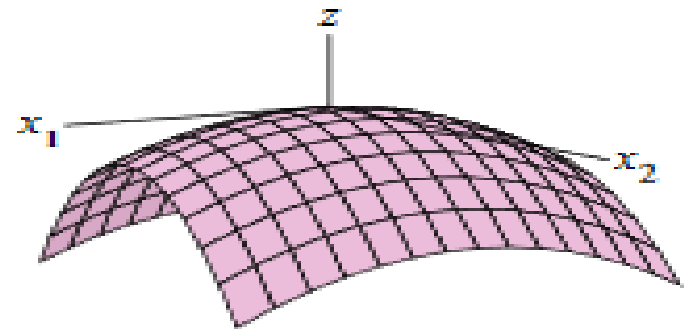
(a) $z = 3x_1^2 + 7x_2^2$



(b) $z = 3x_1^2$



(c) $z = 3x_1^2 - 7x_2^2$



(d) $z = -3x_1^2 - 7x_2^2$

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Graphs of Quadratic forms



Graphically, the graph $z = Q(x)$ is

- convex up if Q is positive definite,
- concave down if Q is negative definite,
- A “saddle” if Q is indefinite.

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EXAMPLE

The quadratic form of $f(x, y) = ax^2 + 2abx + by^2$ may be represented as $x^T A x$ where

$$x = \begin{bmatrix} x \\ y \end{bmatrix} ; \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$



THANK YOU

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